LaTeX Cheat Sheet

Document classes

`article` Best suited for exercise sheets.
Used at the very beginning of a document: `\documentclass{`*class*`}`. Use `\begin{document}` to start contents and `\end{document}` to end the document.

Common `documentclass` options

`10pt/11pt/12pt`	Font size.
`letterpaper/a4paper`	Paper size.
`twocolumn`	Use two columns.
`twoside`	Set margins for two-sided.
`landscape`	Landscape orientation.
`draft`	Double-space lines.

Usage: `\documentclass[`*opt,opt*`]{`*class*`}`.

Recommended packages

`multicol`	Use n columns: `\begin{multicols}{`n`}`.
`mathtools`	Advanced math typesetting. Includes `amsmath`.
`geometry`	Page layout (margins, etc.)
`fancyhdr`	Customize header and footer
`graphicx`	Show image: `\includegraphics[width=`*x*`]{`*file*`}`.
`hyperref`	Insert URL: `\url{`*https://...*`}` or hyperlink: `\href{URL}{text}`.
`enumitem`	Customize `enumerate`.
`tabularx`	Stretchable `X` columns in tables.
`tabulary`	Automatically balanced column widths.
`tabto`	Tab stops (compatible with enumerations).
`beamer`	For creating slides.

Use before `\begin{document}`.
Usage: `\usepackage{`*package*`}`

Title

`\author{`*text*`}`	Author of document.
`\title{`*text*`}`	Title of document.
`\date{`*text*`}`	Date.

These commands go before `\begin{document}`. The declaration `\maketitle` goes at the top of the document.

Miscellaneous

`\pagestyle{empty}`	Empty header, footer and no page numbers.
`\tableofcontents`	Add a table of contents here.

Document structure

`\part{`*title*`}`	`\subsubsection{`*title*`}`
`\chapter{`*title*`}`	`\paragraph{`*title*`}`
`\section{`*title*`}`	`\subparagraph{`*title*`}`
`\subsection{`*title*`}`	

Use a `*`, as in `\section*{`*title*`}`, to not number a particular item—these items will also not appear in the table of contents.

Text environments

`\begin{comment}`	Comment (not printed). Requires `verbatim` package.
`\begin{quote}`	Indented quotation block.
`\begin{quotation}`	Like `quote` with indented paragraphs.
`\begin{verse}`	Quotation block for verse.

Lists

`\begin{enumerate}`	Numbered list.
`\begin{itemize}`	Bulleted list.
`\begin{description}`	Description list.
`\item` *text*	Add an item.
`\item[`*x*`]` *text*	Use *x* instead of normal bullet or number. Required for descriptions.

References

`\label{`*marker*`}`	Set a marker for cross-reference, often of the form `\label{sec:item}`.
`\ref{`*marker*`}`	Give section/body number of marker.
`\pageref{`*marker*`}`	Give page number of marker.
`\footnote{`*text*`}`	Print footnote at bottom of page.

Floating bodies

`\begin{table}[`*place*`]`	Add numbered table.
`\begin{figure}[`*place*`]`	Add numbered figure.
`\begin{equation}[`*place*`]`	Add numbered equation.
`\caption{`*text*`}`	Caption for the body.

The *place* is a list valid placements for the body. `t`=top, `h`=here, `b`=bottom, `p`=separate page, `!`=place even if ugly. Captions and label markers should be within the environment.

Text properties

Font face

Command	*Declaration*	*Effect*
`\textrm{`*text*`}`	`{\rmfamily` *text*`}`	Roman family
`\textsf{`*text*`}`	`{\sffamily` *text*`}`	Sans serif family
`\texttt{`*text*`}`	`{\ttfamily` *text*`}`	`Typewriter family`
`\textmd{`*text*`}`	`{\mdseries` *text*`}`	Medium series
`\textbf{`*text*`}`	`{\bfseries` *text*`}`	**Bold series**
`\textup{`*text*`}`	`{\upshape` *text*`}`	Upright shape
`\textit{`*text*`}`	`{\itshape` *text*`}`	*Italic shape*
`\textsl{`*text*`}`	`{\slshape` *text*`}`	*Slanted shape*
`\textsc{`*text*`}`	`{\scshape` *text*`}`	Small Caps shape
`\emph{`*text*`}`	`{\em` *text*`}`	*Emphasized*
`\textnormal{`*text*`}`	`{\normalfont` *text*`}`	Document font
`\underline{`*text*`}`		Underline

The command (t*ttt*) form handles spacing better than the declaration (t*ttt*) form.

Font size

`\tiny`	tiny	`\Large`	Large
`\scriptsize`	scriptsize	`\LARGE`	LARGE
`\footnotesize`	footnotesize	`\huge`	huge
`\small`	small	`\Huge`	Huge
`\normalsize`	normalsize		
`\large`	large		

These are declarations and should be used in the form `{\small` ...`}`, or without braces to affect the entire document.

Verbatim text

`\begin{verbatim}`	Verbatim environment.
`\begin{verbatim*}`	Spaces are shown as ␣.
`\verb!text!`	Text between the delimiting characters (in this case '`!`') is verbatim.

Justification

Environment	*Declaration*
`\begin{center}`	`\centering`
`\begin{flushleft}`	`\raggedright`
`\begin{flushright}`	`\raggedleft`

Better typesetting (by enabling hyphenation) can be achieved using the package `ragged2e` and its environments `RaggedRight`, `RaggedLeft`, and `Center`.

Miscellaneous

`\linespread{`x`}` changes the line spacing by the multiplier x.

Text-mode symbols

Symbols

&	`\&`	_	`\_`	...	`\ldots`	•	`\textbullet`
$	`\$`	^	`\^{}`	|	`\textbar`	\	`\textbackslash`
%	`\%`	~	`\~{}`	#	`\#`	§	`\S`

Delimiters

' `` ` ``	" ` `` `	{ `\{`	[`[`	(`(`	< `\textless`
' `'`	" `''`	} `\}`	] `]`	) `)`	> `\textgreater`

Dashes

Name	*Source*	*Example*	*Usage*
hyphen	`-`	X-ray	In words.
en-dash	`--`	1–5	Between numbers.
em-dash	`---`	Yes—or no?	Punctuation.

Line and page breaks

`\\`	Begin new line without new paragraph.
`\\*`	Prohibit pagebreak after linebreak.
`\kill`	Don't print current line.
`\pagebreak`	Start new page.
`\columnbreak`	Start new column.
`\noindent`	Do not indent current line.

Spaces and rules

`~`	Fixed space, disallow linebreak (`N.~Biever`).
`\smallskip`	Small vertical space.
`\medskip`	Medium vertical space.
`\bigskip`	Big vertical space.
`\hspace{`l`}`	Horizontal space of length l (Ex: l = `20pt`).
`\vspace{`l`}`	Vertical space of length l.
`\rule{`w`}{`h`}`	Line of width w and height h.

Rules of width/height 0 are sometimes useful for solving spacing or alignment issues.

Tables and tab stops

Tab stops: `tabbing` vs. `tabto`

`\=` Set tab stop. `\>` Go to tab stop.

Tab stops can be set on "invisible" lines with `\kill` at the end of the line. Normally `\\` is used to separate lines.

The `tabbing` environment is incompatible with `enumerate`/`itemize`. In this case it's better to use the package `tabto` instead:

```
\TabPositions{2cm, 5cm}
Some\tab sample\tab text.
```

`tabular` environment

`\begin{array}[`*pos*`]{`*cols*`}`
`\begin{tabular}[`*pos*`]{`*cols*`}`
`\begin{tabular*}{`*width*`}[`*pos*`]{`*cols*`}`

`tabular` column specification

- `l` Left-justified column.
- `c` Centered column.
- `r` Right-justified column.
- `p{`*width*`}` Same as `\parbox[t]{`*width*`}`.
- `@{`*decl*`}` Insert *decl* instead of inter-column space.
- `|` Inserts a vertical line between columns.

`tabular` elements

`\hline`	Horizontal line between rows.
`\cline{`x–y`}`	Horizontal line across columns x through y.
`\multicolumn{`*n*`}{`*cols*`}{`*text*`}`	A cell that spans *n* columns, with *cols* column specification.

Math mode

For inline math, use `\(...\)` or `$...$`. For displayed math, use `\[...\]` or `\begin{equation}`.

2^x	`2^{x}`	x_0	`x_{0}`
$\frac{x}{y}$	`\frac{x}{y}`	$\sqrt[n]{x}$	`\sqrt[n]{x}`
$\sum_{k=1}^n$	`\sum_{k=1}^n`	$\prod_{k=1}^n$	`\prod_{k=1}^n`
$\int_a^b f$	`\int_a^b f`	$\iint_S$	`\iint_S`
$\vec{u}$	`\vec{u}`	$\overrightarrow{AB}$	`\overrightarrow{AB}`

Matrices and determinants:

`\begin{pmatrix}`
`1&2\\ 3&4`
`\end{pmatrix}`
	$\begin{pmatrix}1 & 2\\ 3 & 4\end{pmatrix}$

`\begin{vmatrix}`
`1&2\\ 3&4`
`\end{vmatrix}`
	$\begin{vmatrix}1 & 2\\ 3 & 4\end{vmatrix}$

Math-mode symbols

$\leq$	`\leq`	$\geq$	`\geq`	$\neq$	`\neq`	$\approx$	`\approx`
$\times$	`\times`	$\div$	`\div`	$\pm$	`\pm`	$\cdot$	`\cdot`
$^{\circ}$	`^{\circ}`	$\circ$	`\circ`	$\prime$	`\prime`	$\cdots$	`\cdots`
∞	`\infty`	$\neg$	`\neg`	$\wedge$	`\wedge`	$\vee$	`\vee`
$\supset$	`\supset`	$\forall$	`\forall`	$\in$	`\in`	$\rightarrow$	`\rightarrow`
$\subset$	`\subset`	$\exists$	`\exists`	$\notin$	`\notin`	$\Rightarrow$	`\Rightarrow`
$\cup$	`\cup`	$\cap$	`\cap`	$\mid$	`\mid`	$\Leftrightarrow$	`\Leftrightarrow`
$\dot a$	`\dot a`	$\hat a$	`\hat a`	$\bar a$	`\bar a`	$\tilde a$	`\tilde a`
α	`\alpha`	β	`\beta`	γ	`\gamma`	δ	`\delta`
ϵ	`\epsilon`	ζ	`\zeta`	η	`\eta`	ε	`\varepsilon`
θ	`\theta`	ι	`\iota`	κ	`\kappa`	ϑ	`\vartheta`
λ	`\lambda`	μ	`\mu`	ν	`\nu`	ξ	`\xi`
π	`\pi`	ρ	`\rho`	σ	`\sigma`	τ	`\tau`
υ	`\upsilon`	ϕ	`\phi`	χ	`\chi`	ψ	`\psi`
ω	`\omega`	Γ	`\Gamma`	Δ	`\Delta`	Θ	`\Theta`
Λ	`\Lambda`	Ξ	`\Xi`	Π	`\Pi`	Σ	`\Sigma`
Υ	`\Upsilon`	Φ	`\Phi`	Ψ	`\Psi`	Ω	`\Omega`

Sample LaTeX document

```
\documentclass[11pt]{article}
\title{Template}
\author{Name}
\begin{document}
\maketitle

\section{section}
\subsection*{subsection without number}
text \textbf{bold text} text. Some math: $2+2=5$
\subsection{subsection}
text \emph{emphasized text} text. Display math:
\[
\int_0^1 x^2 \,\mathrm{d}x = \frac{1}{3}
\]

A centered table:
\begin{center}
\begin{tabular}{|l|c|r|}
\hline
first  &  row  &  data \\
second &  row  &  data \\
\hline
\end{tabular}
\end{center}
\end{document}
```


https://wch.github.io/latexsheet/
https://github.com/gschintgen/latexsheet